Aurora-JAX Report

Abstract

Aurora, a foundation model for the Earth system, reports that pretraining beats training from scratch by 25%. It cannot say which of its pretraining datasets earned that gain, because they differ in variables, resolution, physics and volume all at once. We rebuilt Aurora's architecture from primitives in JAX at small scale: 5.1 million parameters, with a spectral-band encoder, a Perceiver level aggregator and a 3-D Swin Transformer U-Net. Its Swin block matches Aurora's own code to 2.8×10^{-7} in float32. We then used it as an instrument on two-dimensional Kolmogorov flow, where every property of the pretraining data can be set independently and ground truth is exact. Twelve staged deliverables, each with an acceptance test, built the solver, data pipeline, model and experiment harness. Nine experiments (237 training runs), each with predictions committed before any run, address the research questions. Dated report follow-ups add 45 registered runs for learning curves and a wider physics gap.

- **Pretraining helps one step ahead.** It roughly halves the one-step error on a held-out regime (42–50%), and the advantage holds out to 2.8× the pretraining band's Reynolds number (36–41%).
- **Proximity tracks zero-shot accuracy.** Neither diversity nor volume changes the fine-tuned advantage at this fixed budget. Distance and optimisation remain confounds in interpreting why.
- **The advantage is a small-data effect.** It falls from 41% with one target trajectory to 5% with six. A well-trained base transfers zero-shot almost as well as a model trained on six target trajectories.
- **The one-step advantage does not carry over to rollouts.** Fine-tuned models drift 51–123% more than scratch over 20 steps.
- **The Perceiver does not earn its place on these mixed-resolution sets.** Pad-and-mask improves the upsampled source's error by 23% originally and by 7–10% with a 100× Reynolds-number gap.
- **Simulations buy no accuracy on real measurements.** On airfoil PIV data, given the same real-data budget, a start pretrained on simulations matches training from scratch.
- **LoRA adapts little.** With 0.84% of the parameters it gains at most about 2% at the unseen viscosities, except where the base model is systematically wrong (71% at Re 30).

Of the original 49 pre-registered experiment predictions, 25 hit; the dated report follow-ups hit 1 of 6. Several misses overturned leads from earlier single-seed stages.

1. Introduction

1.1 The question

Foundation models for physical systems follow a two-step recipe: pretrain one large model on many heterogeneous datasets, then fine-tune it for each task. Aurora [1] is the most prominent example. It is a 1.3-billion-parameter model pretrained on over a million hours of geophysical data, and its pretrained model beats training from scratch by 25%. That one number motivates much of the programme. But the paper cannot say *which part* of the pretraining data earned it. Its source datasets differ in variables, resolution, physics and volume all at once, and rerunning the ablation would cost a pretraining run on 32 A100 GPUs for 150,000 steps.

This project asks the question where it can be answered:

> **Which properties of pretraining data produce the pretraining advantage: volume, diversity, heterogeneity of format, or distance from the target?**

On two-dimensional Kolmogorov flow the data is generated, so each property is an independent knob. The data is also noise-free with exact ground truth, so a validation score measures what it claims to measure.

1.2 Where it came from

The question is the one UGP-I ended on. UGP-I was the author's entry to the NeurIPS 2026 RealPDE competition (Track 1, sim-to-real, airfoil PIV data; development-phase score 80.08). It closed with an evidenced but untestable claim: the remaining gap was a better *pretrained* backbone. The competition banned outside weights and offered only 81 real trajectories. Its offline validation also had no predictive power for the main accuracy channel. On synthetic data none of these obstacles exists. Experiment E6 (§5.5) returns to the competition's data and tests the claim directly.

1.3 Research questions

	Question
RQ 1	**Volume vs diversity.** At a fixed number of training samples, is it better to have many trajectories from a few flow regimes, or a few from many?
RQ 2	**Does the Perceiver encoder earn its place** on deliberately mixed resolutions and variable sets?
RQ 3	**How far does pretraining transfer out of distribution?**
RQ 4	**Which architectural components matter?** Shifted windows, depth, latent level count, the pushforward trick, MAE vs MSE.
RQ 5	**Parameter-efficient adaptation.** Does LoRA match full fine-tuning when fine-tuning data is scarce?

1.4 Contributions

1. **A from-primitives JAX reimplementation of Aurora's architecture** at 5.1 M parameters, written in pure JAX [17] and Optax. Its Swin block matches Aurora's to 2.8×10^{-7} (float32) and its backbone to 3×10^{-16} (float64), given the same weights.

2. **A validated pseudo-spectral Kolmogorov solver and dataset generator.** Energy and enstrophy budgets close to 10^{-9}. The generator is checksummed and resumable, and regenerates bit for bit with single-threaded XLA on a fixed machine.
3. **An ablation harness** that turns one config file into one run and one row of results. It refuses a config without a written prediction, and refuses a held-out regime that leaks into training.
4. **Nine pre-registered experiments** (237 runs) answering RQ1–RQ5, including a sim-to-real test on real measurements and three attention experiments; 45 dated report-follow-up runs supplement them.
5. **Measured answers about the training loop itself:** where its time goes, and JAX against PyTorch.

1.5 Outline

- §2 gives background and §3 the methods.
- §4 reports the twelve stages, §5 what pretraining data buys (E1–E6), and §6 which components matter (E4, E7–E9), with training speed.
- §7 discusses the findings and their limits, and §8 concludes.

---

2. Background

2.1 Aurora's architecture

Aurora [1] is a one-step simulator: given the state at two consecutive times, it predicts the next. It has three parts:

- **Encoder.** Each variable is cut into patches and embedded. Position, patch-area, level and time encodings are added. A Perceiver [2] then maps any number of atmospheric pressure levels onto a fixed number of latent levels. Per-variable embeddings, the patch-area encoding and the Perceiver are what let datasets with different variables, resolutions and level counts share one model.
- **Backbone.** A 3-D Swin Transformer [3] U-Net [4] with windowed and shifted-window attention. Each block is "res-post-norm", with an adaptive LayerNorm conditioned on lead time.
- **Decoder.** It mirrors the encoder.

Aurora trains with a mean-absolute-error (MAE) loss, rolls out autoregressively, and adapts to long rollouts with LoRA [5].

2.2 The testbed: Kolmogorov flow

Two-dimensional incompressible Navier–Stokes in vorticity form, doubly periodic, with sinusoidal forcing and linear drag:

```
∂ω/∂t + u·∇ω = ν∇²ω − αω − A k_f cos(k_f y)
```

The Reynolds number (set through ν) and the forcing wavenumber k_f define a flow regime. The flow is a standard two-dimensional turbulence benchmark [6, 7]. It is genuinely chaotic (Stage 1 measured positive Lyapunov exponents; §4), cheap, and noise-free.

2.3 Related work

Learned simulators for fluids [8] and autoregressive neural PDE solvers [9] motivated the one-step recipe. Brandstetter et al. [9] introduced the pushforward trick tested here. Several recent models pretrain on many PDE families and fine-tune downstream [10–12]. Subramanian et al. [13] characterised how transfer scales with model size and downstream data, how it degrades as physics parameters move out of distribution, and how a model

pretrained on a mixture of physics adapts. This project asks a related question about the pretraining data itself. It differs in method: one PDE family whose data properties can be varied one at a time, an architecture copied from a production foundation model, and predictions fixed before every run.

---

3. Methods

3.1 Solver (Stage 1)

- **Numerics:** pseudo-spectral in space with 2/3-rule dealiasing [14], and Lawson integrating-factor RK4 time stepping [15], all in float64. Energy and enstrophy budgets are tracked at every step.
- **Resolution rule (Stage 2, M1):** a generation grid is accepted when doubling it changes the saved fields by at most 10^{-3} at t = 1. That gives 128^2, 256^2 and 512^2 for Re 100, 300 and 1000.
- **Trajectories:** spin up for 50 time units, then record 100, with a frame every $\Delta t = 0.2$ (501 frames). Frames are saved at 64^2, and for `acc_v1` also at 128^2: velocity u, v, pressure p and vorticity ω.

3.2 Datasets (Stage 2)

- **`acc_v1`:** Re 100, 300 and 1000 at k_f = 4, 8 seeds each, at 64^2 and 128^2. Used by Stages 7–10 and E2, E4 and E5.
- **`e1_v1`:** 326 trajectories over 33 regimes (Re 125–700 × k_f 3–6), at 64^2. Generated on CPU in single-threaded lanes for E1 and E3, and reused as the task for E7–E9.
- **Generator:** resumable, with a SHA-256 manifest. The loader splits by regime and serves windows of frames.
- **Rule:** test sets are always whole held-out regimes or trajectories, never random frames.

3.3 Model (Stages 3–7)

The configuration every experiment uses: 64^2 fields, width D = 64, patch P = 4, L = 3 latent levels and 12 Swin blocks, for **5,106,688 parameters**. Per sample:

Step	What it does	Shape
input	two snapshots, 4 fields each (u, v, p, ω), normalised	2 × (4, 64, 64)
levels	split each field into 5 spectral octave bands with the FFT (these play the role of Aurora's pressure levels; they sum back to the field)	(4, 5, 64, 64)
embed	4 × 4 patches, per-variable linear embedding summed over variables, plus position, patch-area and level encodings; the two snapshots are summed with learned frame vectors	(5, 256, 64)
Perceiver encoder	3 latent queries attend over the 5 level tokens at each position	(3, 256, 64)
backbone	Swin U-Net on a 3 × 16 × 16 token grid: 3 stages down and 3 up (widths 64, 128, 256), 2 blocks each, windows of 3 × 4 × 4, shifted every second block, skips, AdaLN on lead time	(768, 128)
Perceiver decoder and heads	back to 5 levels, per-variable heads, sum over levels	(4, 64, 64) = Δ
output	X(t) + Δ; the heads start at zero, so the untrained model is persistence	(4, 64, 64)

- **Differences from Aurora** [1] (see `docs/architecture.md`): the levels are spectral bands rather than pressure levels; the windows wrap around in both x and y, because the box is periodic; and the model is about 250 times smaller.

- **Extra attention options:** a relative-position bias [3] and q/k normalisation [16] can be switched on; both are off by default and tested in E7–E9.
- **Most of the parameters sit in the backbone's deepest stage** (67%). A forward pass costs about 1.3 GFLOP per sample.
- **History limitation.** The same linear embedding is applied to both frames and then summed. Adding two frame vectors before this sum does not preserve their order: the learned correction sees their sum, while the residual still singles out the current frame. This differs from Aurora's ordered temporal patch projection. Section 8 proposes an explicit ordered-history encoder; no result here uses that proposed change.

3.4 Training and evaluation

- **Training:** one-step MSE loss, Adam with cosine learning-rate decay (to 1% of the peak), batch 16 (batch 8 in Stages 7–10).
- **Precision:** float32 with matmul precision `highest`, that is TF32 off, in every experiment run.
- **One-step error:** the relative L2 error of the predicted next frame on held-out trajectories, often reported as a ratio to persistence (predicting no change).
- **Rollout drift:** the relative error of an autoregressive rollout, averaged over its first 20 or 50 steps.
- **Advantage:** adv = 1 − e(pretrained and fine-tuned) / e(scratch), paired by seed. A fine-tune and the scratch run of the same seed share training trajectories.
- **Reporting:** every cell uses at least 3 seeds, given as mean ± SE.

3.5 Rollout, pushforward and LoRA (Stages 8–10)

- **Rollout:** the model applied to its own output.
- **Pushforward** [9]: training adds a second loss term in which the model's own one-step prediction, with its gradient stopped, is the input. So the model learns from inputs drawn from its own error distribution without backpropagating through the unroll.
- **LoRA** [5]: rank-r factors on every backbone block's attention projections (qkv and output). At r = 4 that is 43,008 trainable parameters, 0.84% of the model.

3.6 Harness and protocol (Stage 11)

The harness. Every run is one JSON config, hashed. The harness refuses a config without a written prediction, and refuses one whose held-out regime appears in training. Each run appends one row recording the full config, its hash, the prediction and the git commit, so every number traces back to one run.

The protocol. It carries over rules paid for in UGP-I:

- Every stage and experiment has a specification in `handouts/` with predictions, committed in git before any measuring run (E1–E5 were frozen on 28 Sep 2026).
- A miss stays a miss.
- Changes after a run are dated amendments, and the original verdict stays beside the new one.
- Splits hold out full trajectories, rather than frames from a training trajectory. Transfer tests also specify which target regimes are excluded from pretraining.
- Any fit is treated as inflated until it is scored on disjoint data.

3.7 Compute

- **Data:** generated mostly on CPUs (a cloud container and the lab machine's Xeon). The lab copy of `acc_v1`'s Re 1000 was generated on a GPU.
- **Early stages:** built and first run on CPUs.
- **E1–E5, Stage 12 and the Stage 10 follow-ups:** the lab's shared GPU machine (4 × NVIDIA RTX PRO 6000 Blackwell, JAX 0.10.2), on idle cards only.

- **E6–E9 and Stage 4's longest follow-ups (F3, F4):** Kaggle's T4 GPUs, used when the lab's cards were busy, with software pinned to the lab machine's versions.
- **Tests:** 125 tests pass on CPU (17 Oct, including seven report-follow-up checks; the original 118-test baseline also passed the 9 Oct audit).

---

4. Results I: the instrument (Stages 1–12)

Each stage has an acceptance test from the original plan and is not passed until that test is met.

#	Deliverable	Acceptance test	Outcome
1	Pseudo-spectral solver	energy budget < 1%; expected inertial-range scaling	**Met.** Budgets close to $\leq 1.7 \times 10^{-9}$; temporal order 3.998–4.015; forward-cascade slope -4.38 ± 0.03 at Re 1000; Lyapunov exponents 0.39 / 0.47 / 0.57 at Re 100 / 300 / 1000; 11 gates pass; 13 of 16 predictions hit
2	Dataset generator and loader	reproducible from seed; spectra stable across trajectories	**Met for single-threaded generation on a fixed machine (F2).** Spectra stable (Q4–Q8 hit). With XLA's default multi-threading a full-length Re 1000 trajectory does not regenerate bit for bit (Q3 and F1 missed)
3	Spectral levels, patch embed and unembed	round trip at machine precision	**Met:** 1.9×10^{-14} in float64
4	Perceiver level aggregator	one model maps C = 3 to 8 levels to L = 3 and back	**Met with 32× the training and a decaying learning rate (F4):** 0.008 / 0.002 / 0.001 / 0.001 at C = 3 / 5 / 6 / 8, against bars 0.02–0.05. Not met at the registered 6000 steps (0.167 at C = 3)
5	Windowed and shifted-window attention	matches a reference Swin block to 10^{-5}	**Met:** matches Aurora's own block to 2.8×10^{-7} (float32), 5.2×10^{-16} (float64)
6	U-Net backbone	matches the reference; overfits one trajectory	**Met:** matches Aurora's backbone to 3.1×10^{-16}; overfits to 0.070× persistence in 16,000 steps (the registered 4000 were too few)
7	Full model	identity at initialisation; trains one-step prediction	**Met:** 0.32× / 0.46× persistence on unseen trajectories at Re 100 / 300 after 3000 steps
8	Rollout and MAE loss	stable 50-step rollout	**Met by the MSE model.** A single-seed MAE model blew up in 3 of 4 Re 300 rollouts; with 3 seeds in E4 this did not reproduce
9	Pushforward	lower drift at matched compute	**Met in E4** (3 seeds, 10,000 steps): −45% / −47% at Re 100 / 300. Not met at Stage 9's own budget (3000 steps, one seed)
10	LoRA fine-tuning	adapts to an unseen ν with < 1% of parameters trainable (error ≤ 0.8× zero-shot)	**Not met.** See §5.3
11	Ablation harness	one config → one run → one row	**Met:** reproduces Stage 7 exactly (difference 0)
12	Training speed	a JAX configuration ≥ 1.5× faster, numerics unchanged	**Met:** 2.07× (§6.4)

Three findings from building the instrument shaped everything after it.

- **Under-training dominates small-budget comparisons.** Stage 6's overfit check missed its step budget and met its target with 4× the steps. Stage 3's learning checks were still improving when their budgets ran out. Stage 4's Perceiver missed every bar at its registered 6000 steps:
    - at 32× the steps with a constant rate, it reached the hardest bar only intermittently;

- with a cosine decay, it met and held all four bars.

  Stage 10 later showed the same effect (§5.3).
- **Chaos turns any last-bit difference into a different trajectory.** At Re 1000 ($\lambda \approx 0.57$) a rounding difference grows by about e^{28} over the 50-unit spin-up.
  - Bitwise regeneration needs identical arithmetic. Follow-up F2 showed that XLA's multi-threaded CPU execution on the 512^2 grid does not give the same bits from run to run, while single-threaded execution does.
  - Two multi-threaded runs already differed by 6.7×10^{-8} after the spin-up. Single-threaded, all three test trajectories, a full-length Re 1000 one included, regenerated identically.
  - Every trajectory is still a valid sample of the dynamics, so the data's statistics are unaffected.
- **Single-seed leads are unreliable.** Stage 8's MAE blow-up and Stage 9's "pushforward does not pay at matched compute" were both overturned by E4's 3-seed, longer-budget runs (§6.1).

---

5. Results II: what pretraining data buys (E1–E6)

5.1 RQ1: volume vs diversity (E1)

Design.

- **Target regime:** (Re 300, k_f 4), held out.
- **Pretraining cells:** n ∈ {2, 8, 32} regimes crossed with T ∈ {32, 64, 128} total trajectories. The regimes come from Re 125–700 × k_f 3–6. Each cell pretrains for 20,000 steps.
- **Fine-tuning:** 2000 steps at lr 1e-4 on the target's trajectories 0–1.
- **Scratch:** trains on the same two trajectories for 22,000 steps.
- **Evaluation:** target trajectories 2–5, with 3 seeds (57 runs).

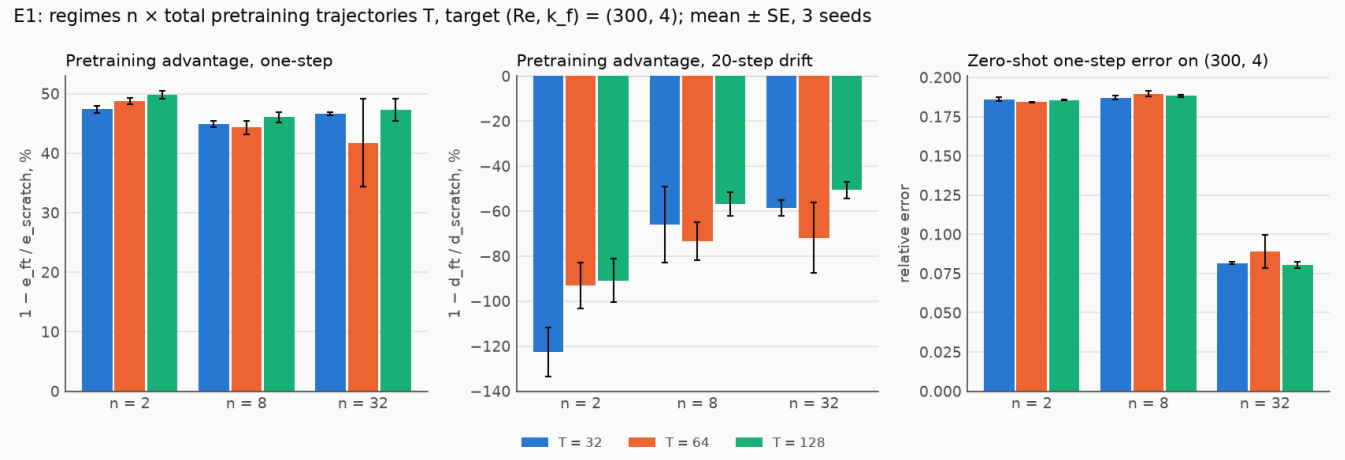
n regimes	adv, one step (over T)	adv, 20-step drift	zero-shot error	fine-tuned error
2	47.4–49.8%	−91 to −123%	0.184–0.186	0.076–0.079
8	44.3–46.0%	−57 to −73%	0.187–0.190	0.081–0.084
32	41.8–47.2%	−51 to −72%	0.080–0.089	0.079–0.087

Scratch reaches 0.150 ± 0.002 (0.49× persistence). Predictions: 2 of 6 hit.

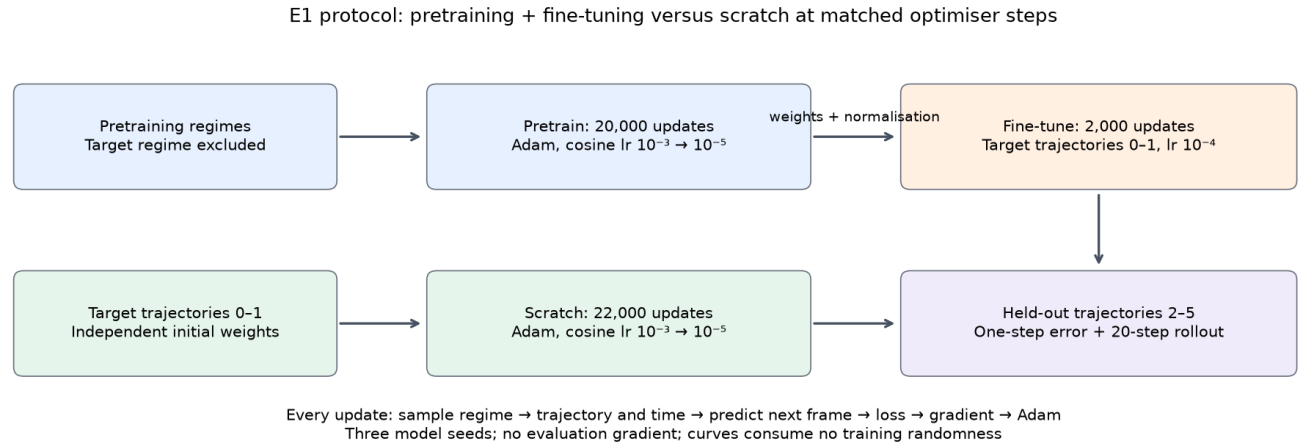
- **Pretraining helps everywhere.** It roughly halves the one-step error in every cell (41.8–49.8%, each at least 5.6 SE).
- **Neither diversity nor volume raises the advantage.** At fixed T, going from 2 to 32 regimes changes it by −3.4 points on average. At fixed n, going from 32 to 128 trajectories changes it by +1.4 points. The cells differ by about their seed spread (E1-P2 and P3 missed).
- **Zero-shot quality follows distance, not regime count.** The n = 32 sets beat the n = 2 and n = 8 sets zero-shot at every T (0.080–0.089 against 0.184–0.190).
  - But n = 2 and n = 8 share the same nearest regimes, (200, 3) and (450, 5), and have the same zero-shot error.
  - The gain appears only at n = 32, the only set with neighbours at the target's k_f = 4: (250, 4) and (350, 4).
  - So the zero-shot effect of "diversity" here is the effect of a nearer neighbour. The n = 2 against n = 8 contrast measures diversity at fixed distance, and finds none.

- **Fine-tuning erases the differences.** After fine-tuning on two target trajectories every cell lands at 0.076–0.087.
- **In rollouts, pretraining hurts.** Fine-tuned models drift 51–123% more than scratch over 20 steps, and the drift ordering of the cells is unrelated to the one-step ordering (Spearman −0.38).

Caveat. Every cell has the same 20,000 pretraining steps, so T = 128 sees each frame four times less often than T = 32. The flat response to volume may reflect this fixed budget.

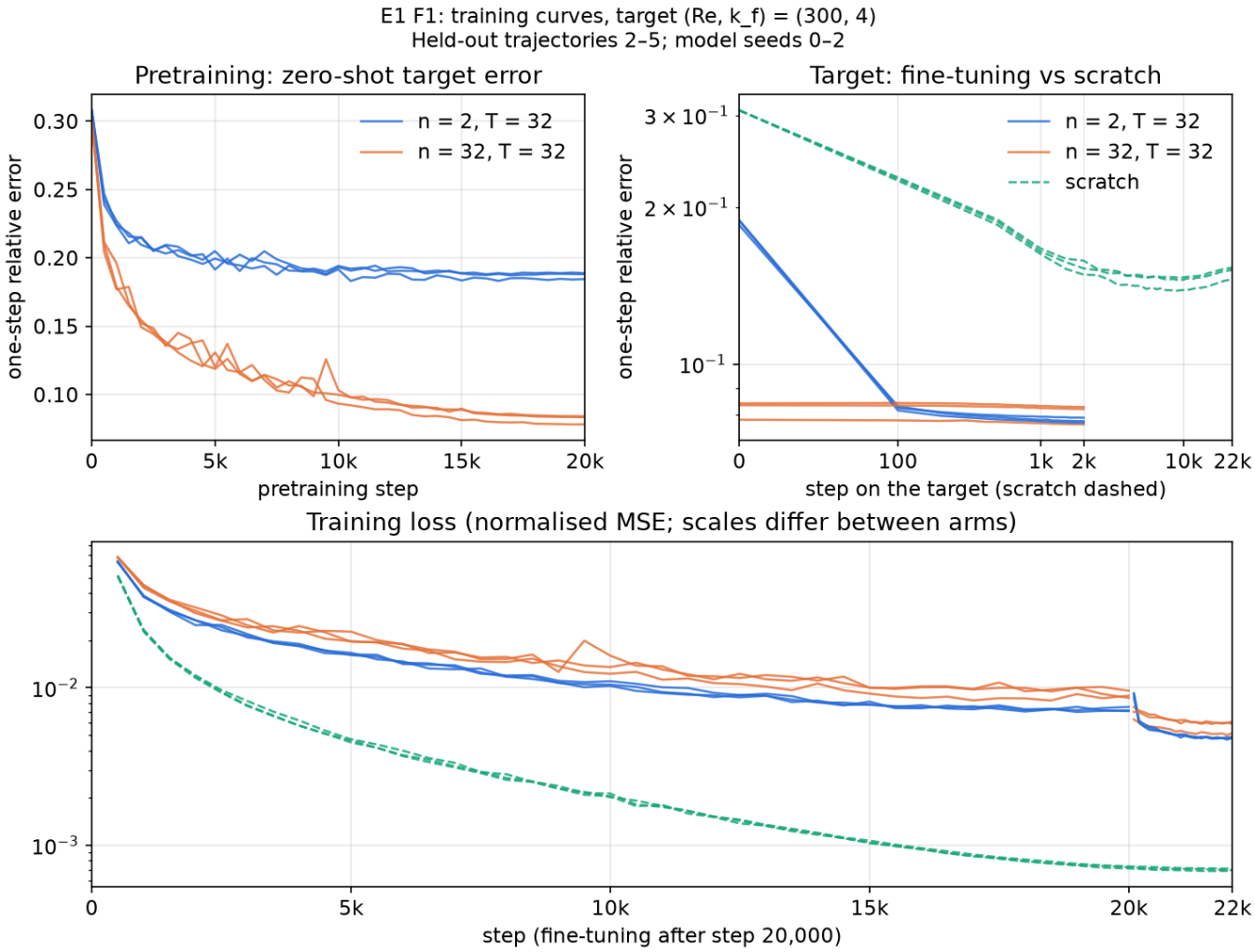


Training protocol and follow-up figures (17 Oct). Pretraining excludes the target regime; fine-tuning and scratch share the two target training trajectories. The evaluation trajectories are never used for a gradient or checkpoint selection.



Learning curves (E1 F1, 15 reruns). The n = 2 and n = 32 cells at T = 32 were rerun with all three seeds, together with scratch, logging the full target evaluation error. These are new training runs, not recovered logs from the original E1 checkpoints. Nine of 15 final errors are within 2% of their originals; the other six differ by 2.15–2.82%, so the registered replication prediction F1-P1 misses. The CPU logging gate leaves parameters bitwise unchanged; that gate does not establish bitwise reproduction of the historical GPU training. The original E1 table and verdicts are retained.

Scratch ends at 0.1499 ± 0.0024, but its lowest logged evaluation errors occur at steps 9,000 / 9,500 / 9,000. Its final error is **4.4–5.0% above those minima**, despite training MSE falling from 0.00218–0.00238 near the minima to 0.00069–0.00071 at the end. Thus F1-P2 also misses: scratch does not end within 2% of its minimum. This is consistent with overfitting the two training trajectories under the fixed recipe. All six fine-tunes end at their lowest logged target error. Their final paired advantages are 48.0 ± 0.9% (n = 2) and 46.3 ± 0.5% (n = 32), consistent with the original finding. Selecting the minimum on these evaluation curves would reuse the scored data; a fair early-stopping comparison needs a separate validation split.

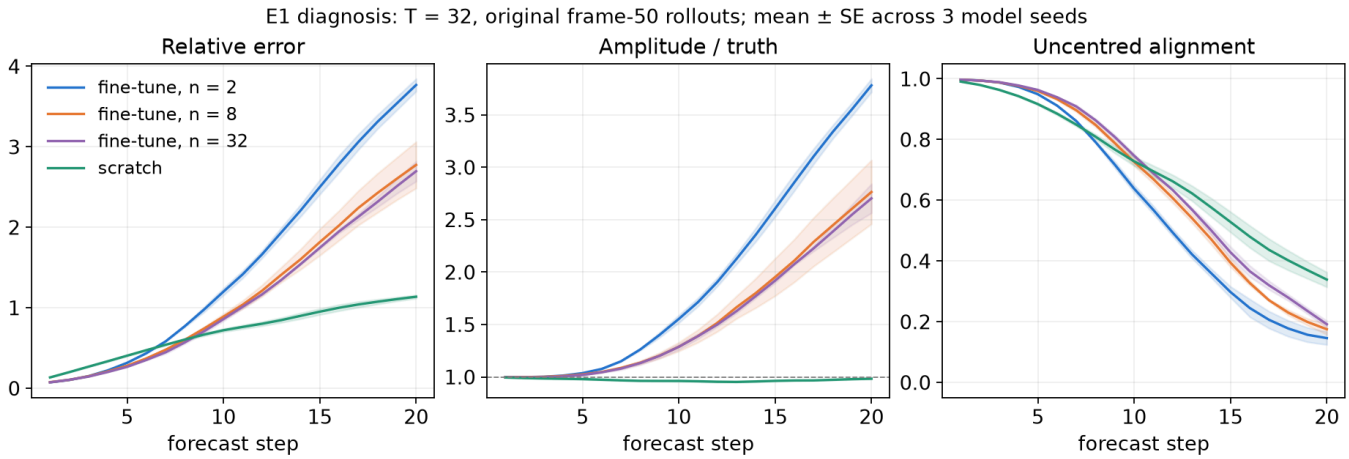


Each line is a model seed. Pretraining / scratch are logged every 500 steps and fine-tuning every 100. Training MSE uses each arm's own normalisation and training distribution, so its values are compared along a curve rather than directly across arms. The target-error panel shows the effect of the short fine-tune and the late rise of scratch's held-out error.

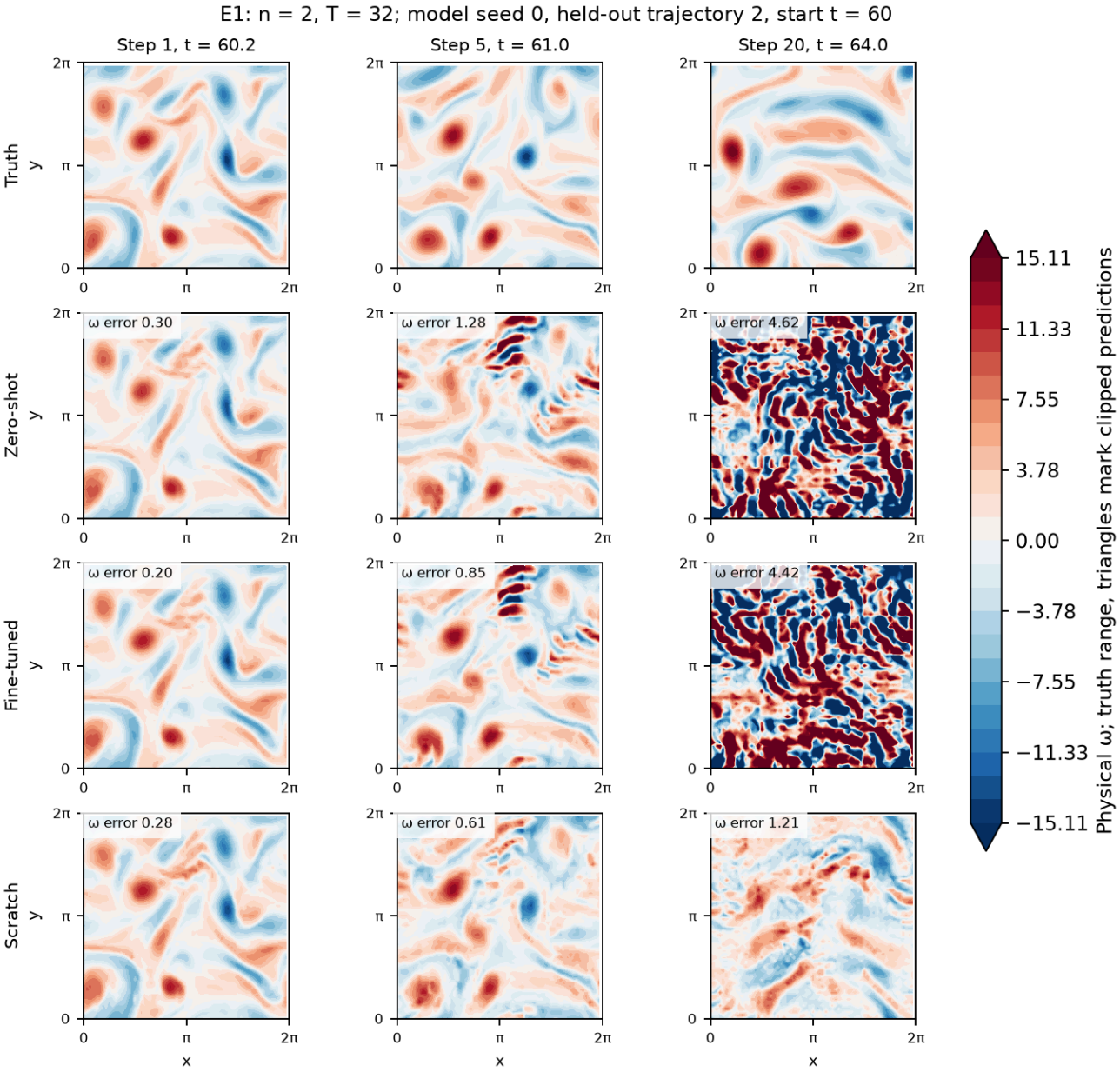
Why does the rollout penalty shrink at n = 8 and n = 32? An exploratory re-evaluation of the original T = 32 checkpoints separates the amplitude from alignment. It uses E1's original four evaluation trajectories, start frame 50, and all three model seeds. Its mean drifts reproduce the original rows within 1.0×10^{-5}; it adds no training.

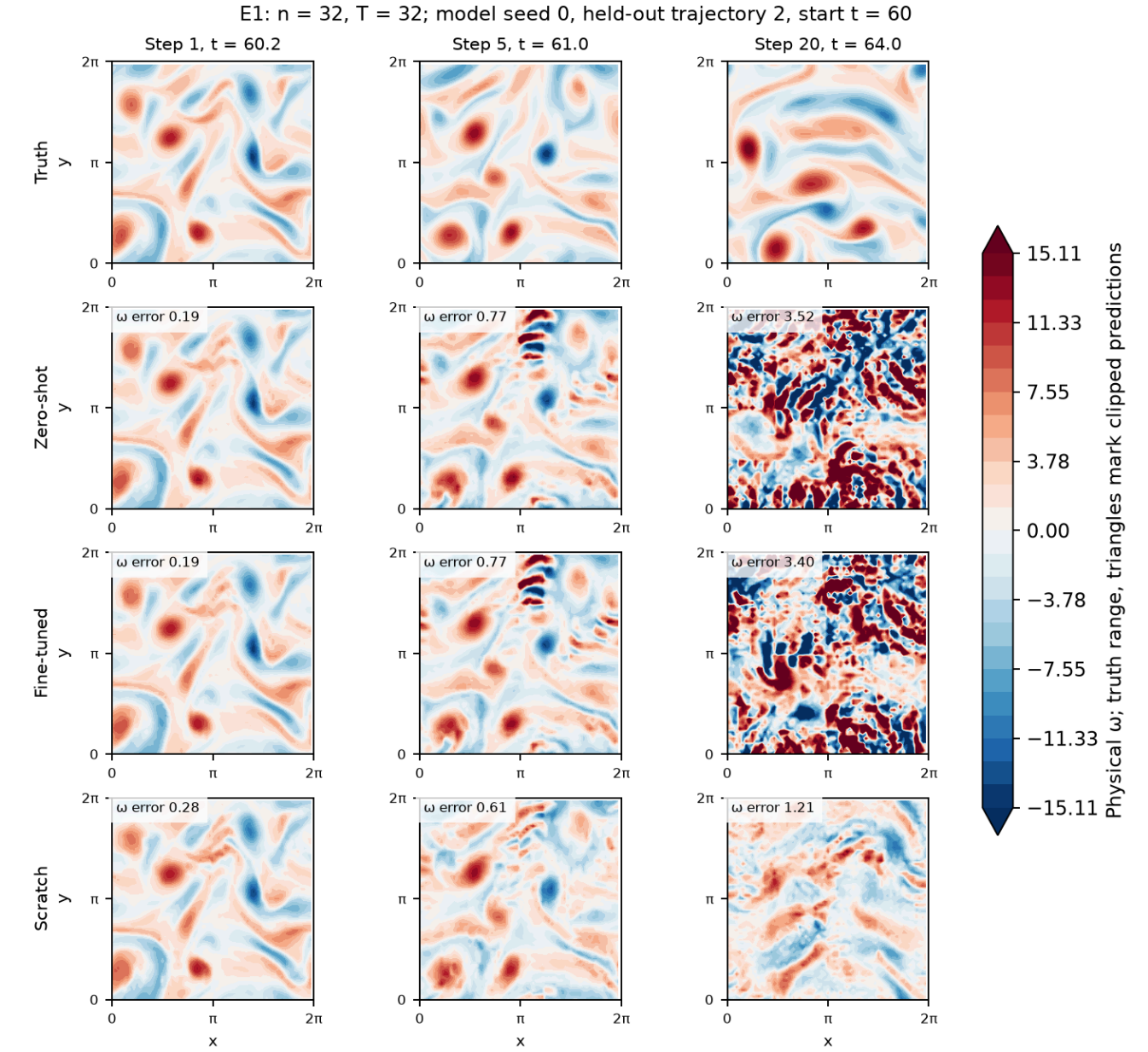
Original E1 arm, T = 32	Mean 20-step drift	Step-20 amplitude / truth	Step-20 alignment
Fine-tuned, n = 2	1.551 ± 0.040	3.782 ± 0.062	0.146 ± 0.021
Fine-tuned, n = 8	1.152 ± 0.085	2.764 ± 0.304	0.175 ± 0.018
Fine-tuned, n = 32	1.107 ± 0.013	2.704 ± 0.139	0.192 ± 0.011
Scratch	0.699 ± 0.019	0.983 ± 0.004	0.339 ± 0.024

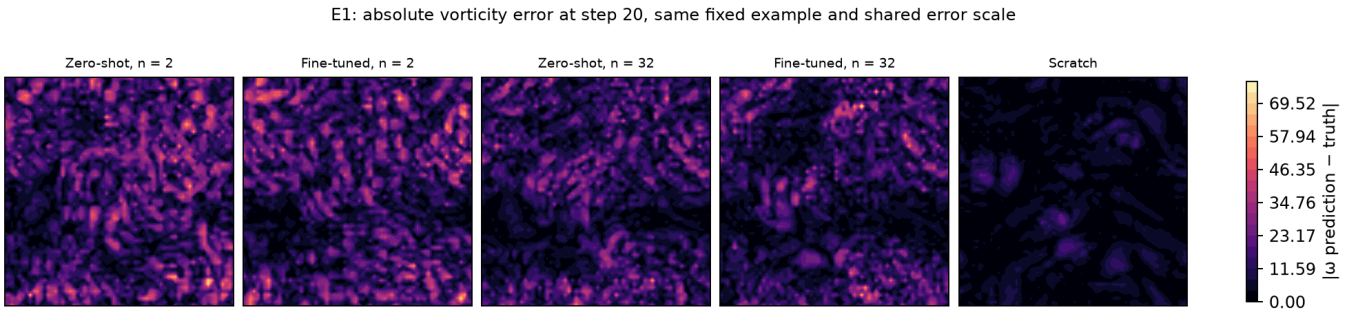
Amplitude a = ‖prediction‖ / ‖truth‖ and alignment c is their uncentred cosine, in each model's training-normalised variables. For each forecast, $e^2 = 1 + a^2 - 2ac$ holds to 1.1×10^{-14}. The n = 8 and n = 32 models grow less strongly than n = 2, which explains much of their smaller late error; their late alignment is also slightly higher. They still grow to about 2.7 times the truth's amplitude, whereas scratch stays near one and retains more alignment. Thus the smaller penalty is not a recovered rollout advantage. One-step fine-tuning does not constrain this autoregressive growth. Why a particular pretraining set changes it is not causally isolated by this diagnostic: n = 32 also changes distance, and the normalisations weight the channels differently. It does not establish that diversity causes stability.



Prediction contours. The following are fixed examples, selected before this re-evaluation: model seed 0, held-out trajectory 2, start frame 50 (physical time 60), forecast steps 1, 5 and 20. They show physical vorticity, using the same truth-based colour range in both figures; colourbar triangles mark predictions beyond that range. The printed vorticity errors use unclipped fields. They are single-example, single-variable errors, distinct from the table's error across all normalised channels. The example illustrates the three-seed diagnosis above; it is not an aggregate score.


E1: n = 2, T = 32; model seed 0, held-out trajectory 2, start t = 60
Step 1, t = 60.2
Step 5, t = 61.0
Step 20, t = 64.0
Truth
Zero-shot
Fine-tuned
Scratch
y
x
ω error 0.30
ω error 1.28
ω error 4.62
ω error 0.20
ω error 0.85
ω error 4.42
ω error 0.28
ω error 0.61
ω error 1.21
0
π
2π
15.11
11.33
7.55
3.78
0.00
−3.78
−7.55
−11.33
−15.11
Physical ω; truth range, triangles mark clipped predictions





5.2 RQ3: out-of-distribution transfer (E3)

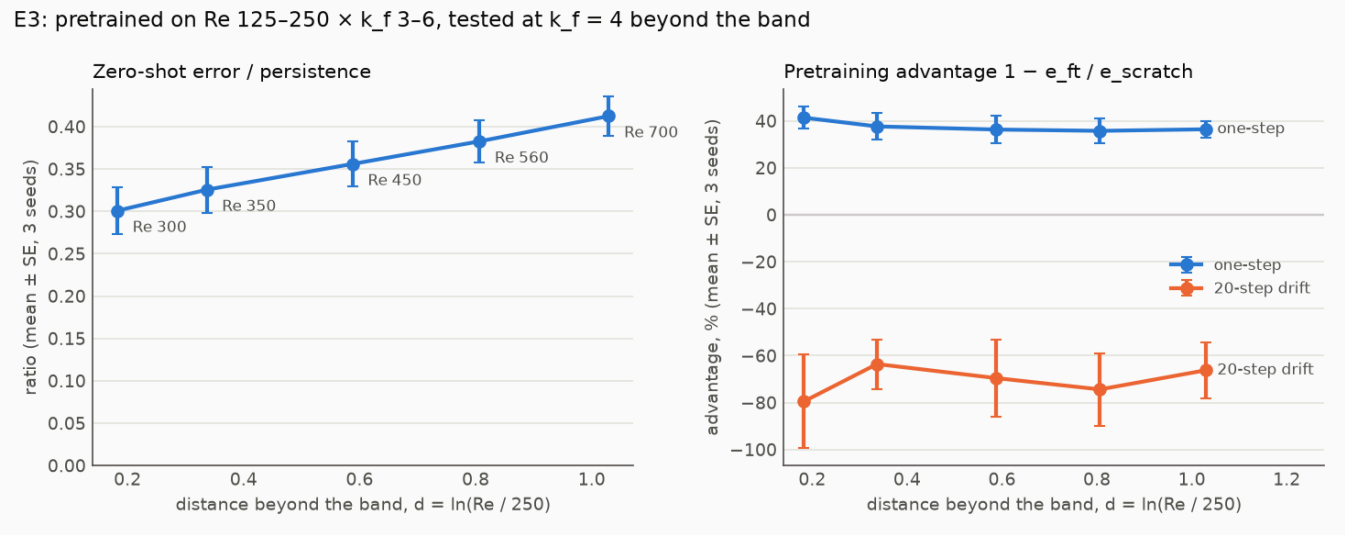
Design.

- **Pretraining band:** Re ∈ {125, 160, 200, 250} × k_f ∈ {3, 4, 5, 6} (64 trajectories, 20,000 steps).
- **Test regimes:** Re 300–700 at k_f = 4, beyond the band. The distance is d = ln(Re / 250), up to 2.8× the band's top Reynolds number.
- **Fine-tuning and scratch:** as in E1 (33 runs).

Test Re	d	zero-shot / persistence	adv, one step	adv, 20-step drift
300	0.18	0.301 ± 0.028	41.4 ± 4.7%	−80 ± 20%
350	0.34	0.326 ± 0.027	37.7 ± 5.7%	−64 ± 11%
450	0.59	0.356 ± 0.026	36.3 ± 5.8%	−70 ± 16%
560	0.81	0.383 ± 0.025	35.8 ± 5.3%	−74 ± 15%
700	1.03	0.412 ± 0.023	36.4 ± 3.5%	−66 ± 12%

Predictions: 3 of 4 hit.

- **Zero-shot degrades slowly and monotonically** with distance (Spearman 1.0). At 2.8× the band's Reynolds number it still beats persistence 2.4-fold.
- **The fine-tuned advantage barely degrades:** 36–41% at every distance. The predicted fall (E3-P3) was not resolved.
- **As in E1, fine-tuned models drift more** than scratch (64–80%).
- **Comparison with E1:** E3's band extrapolates in Re but contains k_f = 4. Zero-shot at Re 300 it reaches 0.30× persistence. That is twice as good as E1's n = 2 and n = 8 sets (0.61×), which enclose Re 300 but lack k_f = 4, and close to E1's n = 32 sets (0.27×).
- **So proximity in k_f matters more than whether the target is enclosed in Re.**



5.3 RQ5: scarce fine-tuning data, LoRA, and Stage 10

E5 design.

- **Pretraining:** Re 100 and 300 for 10,000 steps.
- **Fine-tuning:** on k ∈ {1, 2, 4, 6} trajectories of the unseen Re 1000, for 2000 steps. Either full (lr 1e-4) or LoRA at rank 4 (lr 1e-3).
- **Scratch:** on the same k trajectories for a matched 12,000 steps (39 runs).
- **Zero-shot reference:** the pretrained model's error at Re 1000 is 0.166 ± 0.001.

k	full	LoRA	scratch	gain(full)	gain(LoRA)	adv(k)
1	0.171	0.165	0.290	−3.0 ± 0.4%	+0.6 ± 0.2%	40.9 ± 0.3%
2	0.162	0.164	0.235	+2.3%	+1.3%	30.8 ± 1.4%
4	0.158	0.164	0.189	+4.7%	+1.6%	16.2 ± 0.7%
6	0.157	0.163	0.165	+5.3%	+1.7%	4.9 ± 0.7%

gain = 1 − e / e(zero-shot); adv = 1 − e(full) / e(scratch). Predictions: 2 of 4 hit.

- **The pretraining advantage is a small-data effect.** It falls from 41% with one trajectory to 5% with six. Scratch improves fast with data (0.290 → 0.165), while fine-tuning moves the pretrained model little (0.166 → 0.157).
- **Zero-shot transfer does most of the work.** The pretrained model is within 6% of the best fine-tuned model.
- **LoRA is safe but small.** It gains at most 1.7%. It never hurts, while full fine-tuning overfits one trajectory (−3.0%).

Stage 10 set LoRA's acceptance bar before any of this: with two target trajectories and under 1% of the parameters trainable, LoRA's error must reach ≤ 0.8× zero-shot, a 20% gain. Four pre-registered follow-ups traced why it fails:

Study	Setting	Finding
Stage 10	Re 1000, 2 trajectories, 1000 steps	LoRA gains 0.5%, full fine-tuning 3.0%; the base is already at 0.58× persistence zero-shot
F1	rank sweep	0.2% at r = 1, 1.0% at r = 64 (13.5% of the parameters), against 2.7% for full: rank is not the limit
F2	Re 3000 and Re 30, 3 seeds	at Re 3000 LoRA gains 0.46% and full 3.2%. At Re 30, where the base is worse than persistence (1.92×), LoRA cuts the error by 71% and full by 82%. The base's error at Re 30 is systematic (a predicted change 2.2× too large); at Re 1000–3000 it is a direction error that grows with Re
F3	6 trajectories, longer training	a model trained from scratch beats the bar (−26% at Re 3000, −30% at Re 1000); fine-tuning the base does not (full −12%, LoRA −1%). An exact solver that sees only the 64^2 band predicts at 0.054× persistence, so the information is there
F4	a base trained 10× longer	it halves its in-distribution error and transfers zero-shot almost as well as the scratch models (0.201 vs 0.192 at Re 3000; 0.150 vs 0.149 at Re 1000). Fine-tuned, it beats them by 3.7–8.4%. But nothing gains 20% over its own zero-shot: full fine-tuning on six trajectories 7–9.5%, LoRA 2%

- **With the original base, LoRA cannot make the needed change.** Rank, learning rate and data are all ruled out as the limit.
- **With a well-trained base, there is too little left to adapt.** Its zero-shot error is close to that of a model trained on six target trajectories, so no method gains 20%.
- **LoRA does adapt strongly where the base is systematically wrong** (Re 30). But F2's specification fixed in advance that Re 30 does not decide the acceptance test.
- **Stage 10 therefore stays not met, as a measured negative result.** Its 20% bar mostly measured how under-trained the Stage 7 base was. E5 and F4 agree that once the base is good, fine-tuning data adds little at these viscosities.

5.4 RQ2: heterogeneous data and the Perceiver (E2)

Design.

- **Sources:** A (Re 100) and B (Re 300).
- **Mixed sets:** (a) A at 128^2, B at 64^2; (b) both at 64^2, but B with (u, v) only; (c) both mismatches together.
- **Methods:**

    - the Perceiver: spectral levels with any count, per-variable embeddings;
    - pad-and-mask: upsample to one resolution spectrally, zero-fill missing variables with a presence mask;
    - per-dataset encoders.
- **Reference:** a homogeneous set (h), 3 seeds (30 runs).

Set	Perceiver, B	pad-and-mask, B	per-dataset, B	Perceiver, A	pad-and-mask, A
(a) resolution mixed	0.115	**0.088**	0.123	0.048	**0.039**
(b) variables mixed	**0.065**	0.066	0.079	**0.058**	0.059
(c) both	0.072	**0.056**	0.078	**0.055**	0.059
(h) homogeneous	0.102	—	—	0.042	—

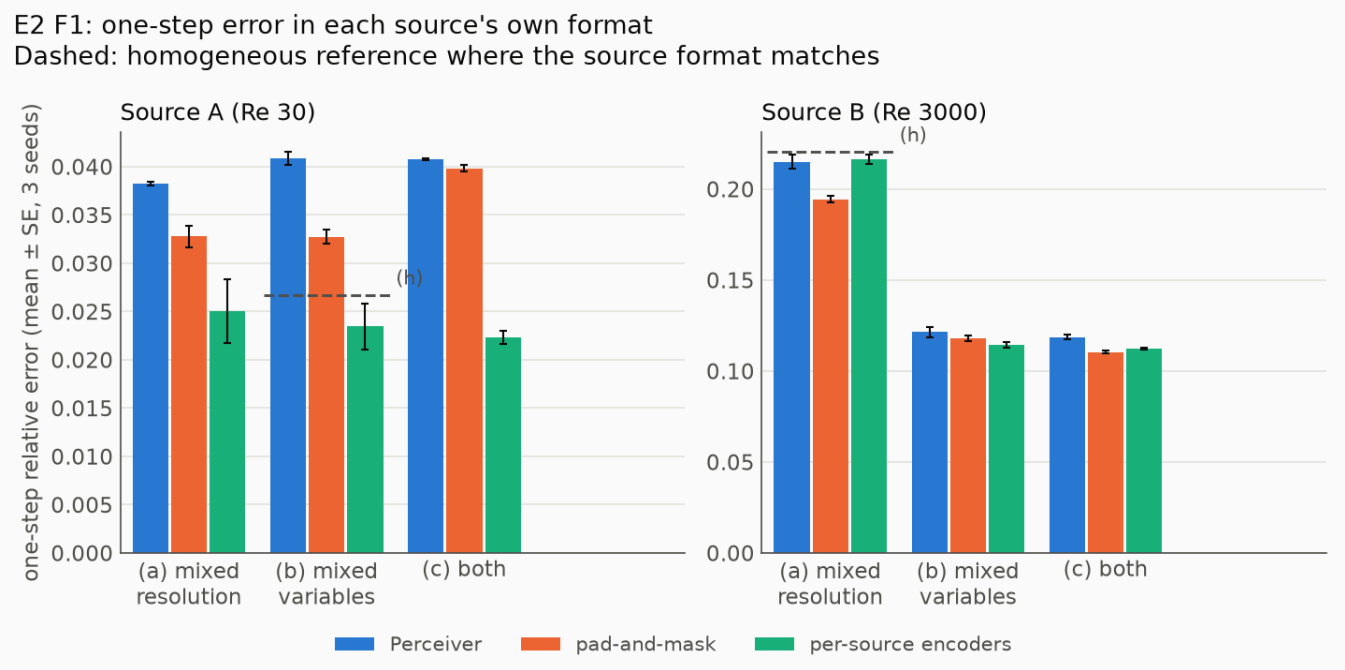
One-step relative error, each source in its own format. Compare within a set: in (b) and (c), B carries only (u, v). Predictions: 1 of 3 hit.

- **Pad-and-mask beats the Perceiver on the upsampled source,** by 23% in (a) and (c).
- **Fusion costs the Perceiver accuracy even where a source's format did not change.** Against (h), its error rises by 13% on B in (a) and by 39% on A in (b): the aggregator does not isolate one source's format from the other's.
- **Per-dataset encoders are the worst** of the three.
- **So at this scale and gap (2× in resolution), the Perceiver does not earn its place.** Whether a larger resolution gap or an unseen format would change this was not tested.

Wider physics gap (E2 F1, 17 Oct). The same 30-run design was repeated with **A = Re 30 and B = Re 3000**, both at k_f = 4. The Reynolds-number gap grows from 3× to 100×, while the methods, formats, 10,000-step training recipes and three model seeds stay fixed. Dataset `e2f1_v1` reuses 12 trajectories and adds four Re 30 trajectories generated on CPU; training seeds 0–5 and evaluation seeds 6–7 are disjoint. Predictions were committed before training. All 30 expected configurations are present exactly once, with finite losses and metrics.

Registered comparison	Relative error change, mean ± SE	Verdict
Pad-and-mask improves B by > 5% in both (a) and (c)	**−9.5 ± 2.3%** (a); **−6.9 ± 1.8%** (c)	HIT
Both unchanged-format Perceiver fusion costs exceed 5%	−2.4 ± 1.7% on B (a); +53.9 ± 7.3% on A (b)	MISS
Per-source encoders are > 5% worse on B in every set	+0.7 ± 2.3% (a); −5.7 ± 2.2% (b); −5.3 ± 1.6% (c)	MISS
Both fusion costs exceed their original values	B: −2.4% versus +12.8%; A: +53.9% versus +38.8%	MISS

Changes mean 100 × (method error / reference error − 1), paired by seed; negative is better. **1 of 4 predictions hit.** The original E2 verdict remains 1 of 3.



The dashed homogeneous reference is comparable on A in (b) and B in (a), where format is unchanged. Compare methods within a set: B has only (u, v) in (b) and (c). **The wider physics gap changes the ranking.** Per-source encoders now lower A's error by 34.4 ± 8.6%, 42.6 ± 5.9% and 45.3 ± 1.6% in (a), (b) and (c), and lower B's mean error in (b) and (c). Pad-and-mask still helps upsampled B, but by less than the original 22–23%. Fusion's cost grows on A and has no resolved positive cost on B; it does not grow uniformly with the physics gap. Specialisation through separate encoders is a possible explanation, not an isolated mechanism. Physics and format remain paired by source. A wider resolution gap and an unseen format remain untested.

5.5 Sim-to-real on real measurements (E6)

Design.

- **Data:** the RealPDE airfoil data. 81 real trajectories of PIV velocity measurements (u, v) at 32 × 64, and 100 simulated trajectories (u, v, p).
- **Test:** the 16 real trajectories at the two lowest and two highest Reynolds numbers.
- **Model:** the E1 model, adapted to rectangular, non-periodic fields.
- **Arms (3 seeds):**
  - **S:** from scratch on the 65 training trajectories, 22,000 steps;
  - **PD and PA:** pretrained on simulations that exclude (PD) or include (PA) the test conditions, then fine-tuned for 2000 steps;
  - **zero-shot** versions of PD and PA, without fine-tuning.
- **Metric:** a 20-step rollout scored by the competition's own scoring code.
- **Input-history difference:** our model uses the last two of the 20 available input frames and predicts recursively. The supplied FNO receives all 20 input frames and outputs the 20-frame forecast together. The targets and evaluation windows match; the information budgets differ.

Arm	Rollout rel_l2	Subscore
S, from scratch	**0.112 ± 0.001**	**94.71**
PD / PA, pretrained, 2000-step fine-tune	0.137 / 0.135	93.58 / 93.68
PD / PA, zero-shot	0.269 / 0.289	88.17 / 87.39
Follow-up F1: PD / PA with S's full 22,000-step budget	0.111 / **0.109**	94.73 / **94.81**
Organisers' FNO, fine-tuned on real data (upper reference)	0.095	95.47

Predictions: 2 of 5 hit, and 3 of 3 in the follow-up.

- **At the registered budget, pretraining hurt:** −22.9% ± 1.5%.
- **Zero-shot transfer barely beats persistence** one step ahead (0.956×).
- **Given the same real-data budget as scratch (F1),** the pretrained starts land where scratch does: +0.4% ± 1.8% (PD) and +2.0% ± 0.9% (PA). So the deficit came from the short fine-tune, and simulation pretraining is a slower route to the same place, not a better one.
- **Simulating the test conditions is worth at most a few percent.**
- **The real data alone go a long way.** Trained from scratch, the model comes within 0.8 subscore points of the organisers' fine-tuned FNO.
- **UGP-I's hypothesis is not supported here.** For this data and model size, a simulation-pretrained backbone does not buy accuracy on the measurements. UGP-I's strong reference model had in fact been fine-tuned on real data (corrected 1 Oct 2026).

Why does the FNO remain ahead? The best matched-budget arm, PAm, has error 0.1094 against 0.0948 for the organisers' fine-tuned FNO: a 15.4% relative error gap, or 0.66 accuracy-subscore points. F1 establishes why the short fine-tune lost to scratch: restoring the real-data budget removes that deficit. It does not identify the cause of the remaining FNO gap. The supplied FNO is an external checkpoint with a different architecture and training history; its real-data fine-tuning may include E6's held-out conditions. It is an upper reference, rather than a controlled architecture comparison. There is also a concrete protocol difference: the FNO sees 20 past frames, against our model's two, and makes a joint forecast instead of 20 recursive calls (`fno_baseline.py`, UGP-I's `r12_eval.window`, and `run_e6.evaluate`). That gives it more temporal information and a different route for accumulating error. History length, the ordered-history limitation (§3.3), boundary treatment and training objective need controlled ablations (§8); their individual contributions to the 15.4% gap have not been measured. In particular, the simulation-only FNO's error is 1.23: the strong FNO reference does not demonstrate that simulation pretraining alone solves transfer to measurements.

5.6 One step ahead vs rollouts

The one-step advantage of pretraining did not carry over to rollouts in any experiment that measured both:

- **E1:** fine-tuned models drifted 51–123% more than scratch over 20 steps.
- **E3:** 64–80% more.

E8 measured rollouts more finely (84 rollouts per model) and showed why one-step and rollout metrics come apart:

- **Every model loses the flow within 15–17 steps.** The correlation between prediction and truth falls below 0.5 there. By step 50 it is 0.07–0.09, and the error is near √2, that of an unrelated field with the right amplitude.
- **So a long-horizon drift measures amplitude more than tracking.** A model that damps amplitude scores better on it without tracking longer (§6.2).

The practical reading: under a short low-learning-rate fine-tune, a report of pretraining on one-step error alone overstates what it buys.

---

6. Results III: which components matter (E4, E7–E9)

6.1 Architecture ablations (E4)

Design. Each variant changes one thing from the base model. The task is Re 100 and 300 (12 trajectories), 10,000 steps, 3 seeds (27 runs). The base reaches 0.192 / 0.330 of persistence one step ahead at Re 100 / 300.

Variant	Δ one-step error (%)	Δ drift, Re 100 (%)	Δ drift, Re 300 (%)
no shifted windows	+6.0 ± 4.5%	+0.8 ± 3.0%	−11.5 ± 6.6%
6 blocks	+9.8 ± 1.6%	−3.4 ± 11.2%	−15.9 ± 9.9%
24 blocks	−6.7 ± 3.1%	+13.4 ± 18.0%	+21.4 ± 27.2%
L = 1 latent level	+2.0 ± 2.8%	+2.4 ± 2.8%	−0.3 ± 6.7%
L = 6 latent levels	−3.9 ± 4.4%	−0.2 ± 14.8%	+0.6 ± 13.9%
pushforward, matched steps	−0.5 ± 2.1%	**−49.8 ± 3.8%**	**−31.3 ± 8.1%**
pushforward, matched compute	+21.5 ± 3.8%	**−44.6 ± 4.6%**	**−47.1 ± 6.9%**
MAE loss	−4.1 ± 3.7%	−19.2 ± 0.8%	−11.1 ± 6.5%

Δ = 100 × (error of variant / error of base − 1), paired by seed. **Negative means lower error (better); positive means higher error (worse).** This is the opposite sign to an advantage or gain, defined as 1 − error / reference in §3.4 and §5.3. Predictions: 1 of 5 hit.

- **Pushforward is the largest effect.** It roughly halves the 50-step drift, even at matched compute (about 10 SE). At matched compute the price is 21.5% one-step accuracy. This overturned Stage 9's single-seed lead.
- **MAE does not blow up.** Across 3 seeds it drifts less than MSE in both regimes, which overturned Stage 8's lead.
- **Depth helps monotonically** (6 → 12 → 24 blocks: +9.8%, 0, −6.7%).
- **Shifted windows matter a little** (+6.0%, 1.3 SE from zero).
- **The latent level count has no measurable effect** on 64² data.

6.2 Pushforward depends on the task (E4 vs E8)

On E1's task (two training trajectories, 22,000 steps), E8 found pushforward to be a trade:

- **It costs one-step accuracy:** 12.9% ± 1.9% worse.
- **It cuts the 50-step drift by only 6.2% ± 1.6%.**
- **Much of that gain is damping.** Its predictions keep 7% less amplitude than the truth from step 10 on. By the identity $e^2 = 1 + \text{amp}^2 - 2 \cdot \text{amp} \cdot \text{corr}$, about two-thirds of its 50-step gain is damping. It does track slightly longer: its correlation falls below 0.5 at step 16 rather than 15.

So pushforward's large drift gain in E4 (12 trajectories, 10,000 steps) does not transfer to a 2-trajectory task.

6.3 Attention inside the windows (E7, E8, E9)

Aurora's Swin attention has no relative-position term, and its queries and keys are unnormalised. On E1's scratch task (3 seeds per arm, paired by seed and batch):

Arm	One-step error	Gain over B	20-step drift gain	Extra training time
B, control	0.150	—	—	—
R, relative-position bias (E7, replicated in E8)	0.140	6.9% ± 0.7%	3.3% ± 0.6%	1–3%
Q, q/k normalisation (E8)	0.139	7.3% ± 0.7%	3.8% ± 1.2%	10%
RQ, both (E9)	**0.136**	**8.9% ± 0.4%**	4.2% ± 1.7%	11%

Predictions: E7 2 of 3, E8 5 of 10, E9 4 of 6.

- **Each change buys about what doubling the depth bought in E4,** at a fraction of the cost.
- **Together they beat either alone in every seed, but they overlap.** Independent gains would give 13.7%; the combination gets 8.9%, about a quarter of the possible extra.
- **The gains fade along a rollout.** They peak at 8–11% at steps 2–3, fall to 1–2% by step 20, and are about zero from step 30.
- **The two act on attention differently (exploratory).**
  - Without q/k normalisation, attention is near-uniform in the fine stages (entropy 0.84–0.93 of the maximum). It is nearly one-hot in the coarsest stage (0.13–0.16), where the logits reach a standard deviation of 8–11.
  - q/k normalisation evens this out across stages. The bias leaves it as it is.

6.4 Training speed (Stage 12)

On the lab card, the E-runs' training loop took 10.2 ms per step with the card only 49% busy: the step is bound by the host, not the card.

Change	Speed-up over the loop
50 steps per jitted call (`lax.scan`), sampling on the device	**2.07×**
buffer donation	1.13×
three seeds in one step (`vmap`)	1.67× per model
bfloat16 matmuls	1.12× (accuracy unchanged)
a fused attention kernel	0.92×

- **Everything that removes host work pays; arithmetic speed-ups barely matter.**
- **Against PyTorch:**
  - eager PyTorch is 2.42× slower than the jitted JAX loop;
  - `torch.compile` is within 11% of it;
  - CUDA graphs make PyTorch 1.36× faster than the loop, but JAX with scan is still 1.52× faster.
- **On a T4 the step is bound by the card instead,** so scan gives only 1.16×.
- **Inside the harness,** scan chunks gave the same models 1.21× faster on a T4 (E8).

---

7. Discussion

7.1 What pretraining data buys

- **The advantage is real.** In every Kolmogorov experiment, a pretrained start cut the one-step error by 31–50% when the target had one or two trajectories (E1, E3, E5).

- **The data property that mattered most was proximity to the target, especially in forcing wavenumber.** Neither the number of regimes nor the number of trajectories moved the advantage at a fixed budget (E1).
- **The advantage shrinks fast as target data grows.** It is 41% with one trajectory and 5% with six (E5).
- **A well-trained base already captures nearly all that fine-tuning would add** at these viscosities (Stage 10 F4).
- **The cost of heterogeneous formats depends on the source regimes.** Pad-and-mask beats the Perceiver on upsampled B in both E2 and its wider-physics-gap repeat. With the wider gap, per-source encoders help and the unchanged-format fusion penalty grows on A but is not resolved on B.
- **Pretraining on simulations of a different flow bought nothing** on real measurements, once the real data got a full training budget (E6).

For the question Aurora leaves open, these results suggest a hypothesis. Much of a pretraining gain may come from pretraining data that lies close to the target. The volume and variety of the pretraining data may matter less. This is a reading of one PDE family at small scale, not a claim about Aurora. In particular, these are fixed-compute results: larger datasets received fewer passes, and the 32-regime set also reduced the nearest-neighbour distance. A distance-matched design with enough optimisation is required before concluding that diversity or volume is intrinsically unhelpful.

7.2 Lessons for evaluating pretrained surrogates

- **Measure rollouts, and separate tracking from amplitude.** One-step error and rollout drift disagreed in every experiment that measured both. A long-horizon drift is dominated by decorrelated steps, where damping looks like skill (§5.6, §6.2).
- **Check the base before judging an adaptation method.** Stage 10's verdict was set mostly by how long the base had trained (F4). Stage 4, and Stage 9 against E4, showed similar budget effects.
- **Use paired, contemporaneous controls.** GPU reruns differed by up to 0.8% in E9; the newly logged E1 reruns differ from their historical endpoints by up to 2.82%. Pair model seeds and report rerun variation when comparing results collected in different sessions.
- **Distance confounds diversity.** Adding regimes usually also brings the nearest one closer. E1 separated the two only because two of its sets share their nearest neighbours.

7.3 The pre-registration record

The original experiments made 49 pre-registered predictions and hit 25. E1–E5, frozen together on 28 Sep 2026, hit 9 of 22. Dated report follow-ups are scored separately in Appendix B. The 17 Oct follow-ups add 45 training runs and hit 1 of 6 predictions: E1 F1 0/2 and E2 F1 1/4.

A miss is not a failed experiment. Every experiment ran as designed, and its result stands. A miss means that the expectation written down before the run was wrong.

- **The predictions were meant to be falsifiable.** Most have numeric bars, for example "the advantage lies in [10%, 40%]" or "at least 9%". A hit rate near one half shows they were genuine bets.
- **Many of the early ones rested on single-seed stage results** that later did not hold.

The 24 misses fall into four kinds:

Kind of miss	Count	Predictions
The expectation was wrong, and the miss is the finding	10	E1-P2 and P3 (diversity did not raise the advantage); E1-P6 (rollouts do not follow one-step error); E2-P1 and P2 (fusion cost the Perceiver accuracy, and pad-and-mask beat it); E6-P1, P2 and P3 (simulation pretraining did not help on real data); E8-P4 and P5 (pushforward on a 2-trajectory task)
Better than predicted	5	E1-P4 (pretraining helped more); E4-P1 (MAE stayed stable); E4-P2 (pushforward paid even at matched compute); E4-P4 (24 blocks helped); E8-P7 (q/k normalisation helped)
Missed by a hair, or within noise	5	E8-P2 (3.3% against a 3% limit); E8-P8 (10.1% against 10%); E9-P1 (8.9% against 9%); E9-P6 (0.8% against 0.5%); E4-P5 (L = 1 at +2.0 ± 2.8% and L = 6 at −3.9 ± 4.4%, within noise but outside the stated bands)
Too noisy to resolve, ill-posed, or smaller than predicted	4	E3-P3 (a fall with distance, not resolved); E5-P2 (5.3% against ≥ 8%); E5-P3 (a ratio that is ill-posed with one trajectory); E7-P2 (4 rollouts could not resolve the drift; E8's 84 did)

Because each miss was recorded against a prediction made in advance, none could be explained away after the fact.

7.4 Limitations

- **Scale and scope.** The model has 5.1 M parameters and sees 64² fields of one PDE family. Effects that need scale or genuinely different physics cannot appear here.
- **Fixed budgets.** E1 held pretraining compute fixed across volumes, and E4 trained for 10,000 steps. Stage 4 and Stage 10 showed that under-training can dominate a comparison, so the volume and depth conclusions hold at these budgets only. E1 F1 also shows that scratch's evaluation error rises after steps 9,000–9,500 while training loss falls. The final-step comparison is reproducible at the level of the finding, but a stopping recipe can affect its size.
- **One fine-tuning recipe.** Most transfer results use 2000 steps at lr 1e-4. E6 F1 and Stage 10 F4 show that the recipe can change the verdict.
- **Three seeds.** Effects of a few percent sit near the resolution of 3 seeds and the observed rerun variation (up to 0.8% in E9 and 2.82% in the later E1 reruns).
- **Confounds left open.** In E1, the n = 32 cells are confounded with distance. E3 extrapolates upward in Re only. E2 tests a 2× resolution gap and no unseen format.
- **Data copies.** The original Stages 7–10 used a cloud copy of `acc_v1`. E2, E4, E5 and the later Stage 10 follow-ups used the lab machine's copy. The copies are statistically equivalent but not bitwise identical, so absolute numbers across those stages carry a small caveat. Comparisons within an experiment do not.

---

8. Conclusion and future work

Conclusion. We built a verified small-scale copy of Aurora's architecture and used it to take apart the pretraining advantage on a flow where the data can be controlled exactly.

- **The advantage is large when target data is scarce, and fades as it grows.** Zero-shot accuracy tracks proximity to the target; diversity and volume do not change the fine-tuned advantage at this budget. Distance-matched, better-optimised controls are needed to separate their causes.
- **It does not survive into rollouts** under a short fine-tune.
- **Among architectural components:**
  - pushforward helps rollouts on a 12-trajectory task, and mostly damps amplitude on a 2-trajectory task;
  - two cheap attention changes each buy 7% one step ahead;
  - the Perceiver, the component meant for heterogeneous data, did not beat a simple baseline.

- **LoRA adapts strongly only where the base model is systematically wrong.**

Future work. Each of these would get its own pre-registered specification.

- **E1:** a 32-regime control matched in distance to the 2-regime set, and pretraining steps scaled with volume. Together they would separate diversity, distance and compute.
- **E3:** a downward-in-Re arm, and an arm at distance in k_f, which E1 suggests matters most.
- **E2:** a wider resolution gap and an unseen format at test time.
- **E4:** longer budgets, since the pushforward and depth effects may change at convergence.
- **Rollouts:** a measure that separates tracking from amplitude. Also fine-tuning recipes that do not trade rollout stability for one-step accuracy, for example pushforward during fine-tuning.
- **Stage 10:** a re-specified test framed relative to a well-trained base, if the question is still of interest.

8.1 Proposed architecture: preserve time order and resolve local measured flow

The first change I would test is an **ordered-history encoder**. Replace the shared-embedding sum by separate projections of $X(t)$ and $X(t) - X(t-\Delta t)$, or concatenate the two frames before a learned temporal projection. Retain the current-frame residual. This gives the backbone access to the direction of the recent change, which the current symmetric token sum discards. It is a small, specific change to test before adding depth. A temporal-swap gate should verify that the learned correction can distinguish the two input orders. A separate history-length ablation should then compare two frames with all 20 available frames, matching the FNO's information budget before attributing its accuracy advantage to the backbone.

For the RealPDE foil measurements, the next proposal is a **boundary-aware local residual branch** beside the Swin backbone. A small spatial convolutional encoder/decoder would operate on the ordered velocity inputs with non-periodic padding and exchange features at each U-Net scale. Its output would add a learned correction to the Swin prediction, with its last projection initialised to zero. This provides a direct path for local shear and wake structure at 32×64, which is otherwise patchified and compressed through spectral bands. Compare this with replacing FFT bands by a spatial multiscale pyramid, since masked Swin windows do not remove the periodic extension implicit in an FFT. These are proposed inductive biases; the existing experiments do not measure their gains.

Two measured lessons constrain this design. First, relative-position bias and q/k normalisation are inexpensive attention options with a joint 8.9% one-step gain on Kolmogorov flow, but no demonstrated long-horizon gain (§6.3); test them on real data rather than assuming transfer. Second, UGP-I's PIV data are a two-dimensional slice of three-dimensional flow. About 45% of their fluctuation energy is non-solenoidal, and hard divergence-free projection lost nine competition points (`project_memory.md`, §2). The proposed velocity head therefore keeps both components unconstrained. A local branch must not enforce a two-dimensional incompressibility condition that the measurements violate.

Evaluation plan, not a new result. First ablate ordered history alone; then the local branch alone and their combination, with attention options in a separate comparison. Use E6's disjoint Re split, at least three seeds, the same real-data optimiser steps and separately matched wall compute, and report parameter count and inference cost. Pre-register the tests before training. Report one-step error, the competition's 20-step rollout metric, amplitude and alignment, plus velocity contours and error maps. A rollout loss should be a separate ablation: E4 and E8 show that pushforward can improve a drift score through damping. No gain is claimed until these controlled comparisons succeed.

Acknowledgements

I thank Prof. Rajesh Ranjan for supervising the project, and Prashant Kumar for proposing the question and the approach. Computations ran on the lab's shared GPU machine and on Kaggle's T4 notebooks.

References

1. C. Bodnar et al., A foundation model for the Earth system, *Nature* 641, 1180–1187 (2025). arXiv:2405.13063.
2. A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, J. Carreira, Perceiver: General perception with iterative attention, *ICML*, PMLR 139, 4651–4664 (2021).
3. Z. Liu et al., Swin Transformer: Hierarchical vision transformer using shifted windows, *ICCV* (2021).
4. O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional networks for biomedical image segmentation, *MICCAI*, LNCS 9351, 234–241 (2015).
5. E. J. Hu et al., LoRA: Low-rank adaptation of large language models, *ICLR* (2022).
6. G. Boffetta, R. E. Ecke, Two-dimensional turbulence, *Annual Review of Fluid Mechanics* 44, 427–451 (2012).
7. G. J. Chandler, R. R. Kerswell, Invariant recurrent solutions embedded in a turbulent two-dimensional Kolmogorov flow, *Journal of Fluid Mechanics* 722, 554–595 (2013).
8. D. Kochkov et al., Machine learning–accelerated computational fluid dynamics, *PNAS* 118 (21), e2101784118 (2021).
9. J. Brandstetter, D. Worrall, M. Welling, Message passing neural PDE solvers, *ICLR* (2022).
10. M. McCabe et al., Multiple physics pretraining for spatiotemporal surrogate models, *NeurIPS* (2024).
11. Z. Hao et al., DPOT: Auto-regressive denoising operator transformer for large-scale PDE pre-training, *ICML* (2024).
12. M. Herde et al., Poseidon: Efficient foundation models for PDEs, *NeurIPS* (2024).
13. S. Subramanian et al., Towards foundation models for scientific machine learning: Characterizing scaling and transfer behavior, *NeurIPS* (2023).
14. S. A. Orszag, On the elimination of aliasing in finite-difference schemes by filtering high-wavenumber components, *Journal of the Atmospheric Sciences* 28, 1074 (1971).
15. J. D. Lawson, Generalized Runge–Kutta processes for stable systems with large Lipschitz constants, *SIAM Journal on Numerical Analysis* 4, 372–380 (1967).
16. M. Dehghani et al., Scaling vision transformers to 22 billion parameters, *ICML* (2023).
17. J. Bradbury et al., JAX: composable transformations of Python+NumPy programs (2018). https://github.com/jax-ml/jax

Appendix A: where each result lives

Topic	Report	Specification
Stages 1–12	`experiments/stageN/REPORT.md`	`handouts/stageNN_*.md`
E1–E9	`experiments/eN/REPORT.md`	`handouts/eN_*.md`
Model walkthrough, shapes, parameters	`docs/architecture.md`	—
Decision record and protocol rules	`project_memory.md`	—
Original plan	`PROJECT_PLAN.md`	—

To check the code and reproduce a result:

```
python3.11 -m venv .venv
.venv/bin/pip install jax numpy scipy sympy pytest matplotlib optax
# correctness gates (125 tests; run on CPU)
.venv/bin/python -m pytest tests -q
# one config -> one run -> one row: reproduces the Stage 7 result
.venv/bin/python experiments/run.py experiments/configs/h1_stage7_repro.json
```

Each report ends with the exact commands that produced its numbers.

Appendix B: prediction record

Study	Predictions hit	Study	Predictions hit
E1 volume vs diversity	2 / 6	E6 sim-to-real	2 / 5
E2 heterogeneous fusion	1 / 3	E6 follow-up F1	3 / 3
E3 out-of-distribution	3 / 4	E7 relative-position bias	2 / 3
E4 ablations	1 / 5	E8 follow-ups	5 / 10
E5 LoRA vs full	2 / 4	E9 bias and q/k normalisation	4 / 6
Experiments, total	**25 / 49**		

Report follow-ups registered on 17 Oct (45 additional runs; 282 including the original study):

Follow-up	Predictions hit	Finding
E1 F1: learning curves	0 / 2	Historical endpoints vary by up to 2.82%; scratch ends 4.4–5.0% above its logged minima
E2 F1: Re 30 / 3000	1 / 4	Pad-and-mask still helps B; per-source encoders help, and fusion cost does not grow uniformly
Report follow-ups, total	**1 / 6**	Original predictions and verdicts retained

Stage follow-ups:

- **Stage 4:** F1 1 / 2, F2 1 / 2, F3 3 / 4, F4 4 / 4.
- **Stage 10:** F1 2 / 2, F2 6 / 6, F3 6 / 7, F4 5 / 6.
- **Stage 2:** F1 0 / 1, F2 3 / 3.